2025 Fall MAT2001 Final Exam

December 10, 2025

Q1. Find the orthogonal trajectories of the given family of curves: $y = ce^x$.

Q2. Give two linearly independent solution of the following second order differential equation:

$$y'' + 2ty' - \lambda y = 0$$

where $\lambda \neq 2n$ for all $n \in \mathbb{N}$.

Q3. Determine all vectors $\mathbf{x}^0$ such that the solution of the initial-value problem

$$\dot{\mathbf{x}} = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 1 & -1 & -1 \end{pmatrix} \mathbf{x}, \quad \mathbf{x}(0) = \mathbf{x}^0$$

is a periodic function of time.

Q4. Let the functions $u(t)$ and $v(t)$ be continuous on the interval $[0, t]$, and let $v(t) \geq 0$. Suppose there exists a constant $K \geq 0$ such that for any t, the following inequality holds:

$$u(t) \leq K + \int_0^t v(s)u(s)\,ds.$$

Show that for any $t \in [0, b]$, the following inequality holds:

$$u(t) \leq K \cdot e^{\int_0^t v(s)\,ds}.$$

Q5. **(i)** Given the following second-order linear homogeneous ordinary differential equations:

$$y'' + r(x)y = 0, \quad z'' + q(x)z = 0$$

where $r(x) > q(x)$ holds for all x in the domain of definition, and y, z are non-trivial solutions of the respective equations. Show that between any two consecutive zeros of z, there exists at least one zero of y.

(ii) Consider the following second-order linear homogeneous ordinary differential equation:

$$y'' + \left(1 + \frac{\sin^2 x}{1 + x^2}\right) y = 0$$

Show that, the function $y(x)$ has infinitely many zeros. Furthermore, when $x \to \infty$, the distance between any two consecutive zeros of $y(x)$ tends to π.

Q6. Consider ordinary differential equation:

$$y^2 - x^2 + \frac{x^4}{2} = c^2$$

Determine whether there exist non-periodic solutions to this equation.

Q7. Consider the following system of ordinary differential equations:

$$\begin{cases} \dot{x} = x\,(1 - x - 2y) \\ \dot{y} = -y\,(1 - 2x - y) \end{cases}$$

(i) Find all equilibrium points of the system, and determine the stability of each equilibrium point.

(ii) Does there exist any globally stable equilibrium point in the system?

Q8. (1) Given the following system of differential equations:

$$\begin{cases} \dot{x} = x\left(e^{x^2+y^2} - 3\right) - y \\ \dot{y} = y\left(e^{x^2+y^2} - 3\right) + x \end{cases}$$

determine whether there exist non-trivial periodic solutions to the system.

(2) Consider the following second-order ordinary differential equation:

$$\ddot{z} + \dot{z} + (\dot{z})^5 - 3z^2 = 0$$

Prove or disprove: whether there exist infinitely many periodic solution.

Q9. Consider the system of ordinary differential equations:

$$\begin{cases} \dot{x} = y + x \cdot \frac{(r-3)^2(r^2-5r+4)}{r} \\ \dot{y} = -x + y \cdot \frac{(r-3)^2(r^2-5r+4)}{r} \end{cases}$$

where $r^2 = x^2 + y^2$. Determine all limit cycles of the system, and discuss the stability of each limit cycle.

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